

Homework 5

Question 1. Please read chapter 8 of Chartrand et al. and write a couple sentences about a topic/example/concept that you found difficult or interesting and why?

I liked the part in 8.4 where it talks about where this meaning of the word combination comes from and Pascals letter. I like learning about the history of math.

Question 2. Each student at a certain university is given a 6-digit code (such as 123789 or 001122).

- (a) How many different codes are there? 10^6 since there are 6 choices of 10 digits per position
- (b) How many codes read the same forward and backward? 10^3 since both halves must match then that narrows our positions to half
- (c) How many codes contain only odd digits? 5^6 since 1,3,5,7,9 are the only odd $\mathbb{N}$ under 10
- (d) How many codes contain at least one even digit? $10^6 - 5^6$ since we already found all codes that are all odds than we can just get all other codes

Question 3. How many different 10-bit strings begin with 1011 or 0110? there are 2^6 combinations per beginning since the first 4 positions are already decided. For both it would be $s(2^6)$

Question 4. A password on a computer system consists of four characters, each of which is either a digit or a letter of the alphabet. Suppose that each password must contain at least one digit and at least one letter. How many different such passwords are there? $36^4 - 26^4 - 10^4$ Since without character constraints it is 36^4 but there cannot be a case where all are letters (-26^4) nor a case where all are digits (-10^4)

Question 5. A total of 70 students who go to football, basketball or hockey games on a regular basis are surveyed as to which of these three events they attend. They responded:

- 38 students go to football games.
- 38 students go to basketball games.
- 35 students go to hockey games.
- 17 students go to both football and basketball games.
- 15 students go to both football and hockey games.
- 16 students go to both basketball and hockey games.

How many go to all three?

7 since there are a total of 70 people but there are 111 games attendended, 48 (17+15+16) of those games are the result of people going to two seperate games or more. That means there are 41 (111-70) games being attended by someone that has attended 1 game before, but there are 48 games that are the result of repeat game attendences which gives us 7 (48-41) games that must have been attended as a third game.

Question 6. How many people must be present to guarantee that

- (a) at least two have the same birthday?
- (b) at least two of their birthdays are in the same month?
- (c) at least three of their birthdays are in one of the months January, February, March, April or at least four of their birthdays are in one of the remaining months?
- (a) You would need 366 since either the last birthday would fall on another or if not, then there is at least one day with multiple birthdays
- (b) You would need 13 people to guarantee that at least 2 birthdays are in the same month since there a 12 months, a thirteenth person would have to be the second in a month or if not then there is already another month that already has 2 or more birthdays in it

- (c) You would need 33 people to guarantee that either condition would be met. If we took the maxes of each condition without being a guarantee we would get $8 + 24 = 32$. An extra person (33) would guarantee that either condition would or would already be met.

Question 7. How many different 8-bit strings contain exactly 5 1s?

56 since $\frac{8!}{5!(8-5)!} = 56$

Question 8. Show that $\binom{2n}{2} = 2\binom{n}{2} + n^2$.

Question 9. A student committee is to consist of 3 seniors and 4 juniors. A total of 6 seniors and 8 juniors have volunteered to serve on the committee. How many committees are possible?

Question 10. A total of 5 seniors, 3 juniors and 4 sophomores have volunteered to serve on a 4-person committee. How many committees are possible if

- (a) there is no other restriction on membership for the committee?
- (b) at least one senior, one junior and one sophomore must serve on the committee?
- (c) at least 3 seniors must serve on the committee?
- (d) at least one senior must serve on the committee?